

Project Design Phase

Solution Architecture

Date:	
Name:	Krishnakumar S
Project Name:	Automated Network Request Management System
Maximum Marks:	4 Marks

Solution Architecture Description

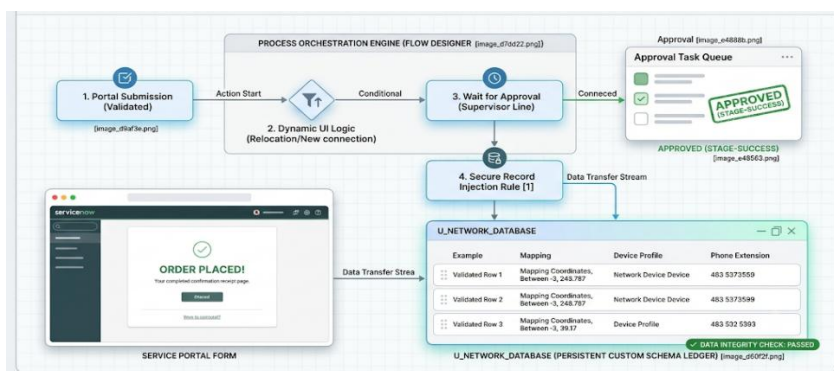
The **Solution Architecture** provides a clear blueprint for transforming the manual network request process into an automated and secure ServiceNow-based solution. It defines how data flows from the self-service portal through approval workflows to the backend database while ensuring accuracy, transparency, and security.

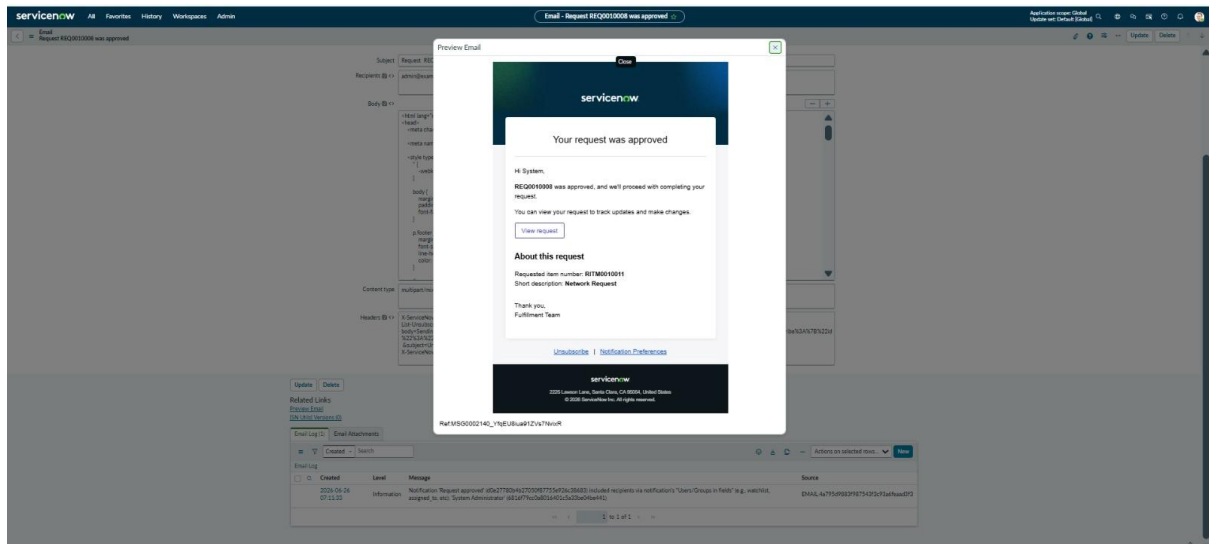
Key Objectives:

- **Reduce Business Challenges:** Eliminate manual email communication, repetitive data entry, and delays in network provisioning.
- **Improve Process Visibility:** Clearly define data flow, approval stages, validation rules, and storage mechanisms for all stakeholders.
- **Establish System Requirements:** Integrate the Service Portal, Flow Designer workflows, and backend database into a seamless automated process.
- **Ensure Security & Data Integrity:** Enforce user authentication, role-based access, input validation, and secure data storage to maintain reliable system records.

Solution Architecture and Data Flow Diagram

- Below is the end-to-end cloud-native architecture mapping how user intake form inputs transform into structured, transaction-verified operational records within the persistent ledger.





Architectural Data Flow Breakdown

The solution follows a four-stage architecture to automate network request processing while ensuring data accuracy, approval control, and secure record management.

1. Portal Submission (Front-End Layer)

Employees submit requests through the Service Portal. The system automatically fills in user details, validates mandatory fields, and dynamically displays relevant fields based on the selected request type (New Connection or Relocation).

2. Workflow Processing (Flow Designer)

After submission, the request is routed to Flow Designer, which manages the workflow, processes the request, and coordinates the next steps without requiring manual intervention.

3. Manager Approval

The workflow sends the request to the appropriate manager for approval. Notifications are generated automatically, and the request proceeds only after the required approval is received.

4. Secure Data Storage

Once approved, the system validates the request data and stores it in the custom `u_network_database` table. The user is then notified with a confirmation message indicating that the request has been successfully submitted and processed.